TabuLLM: Comprehensive Augmentation Audit

1. Executive Summary

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference.

Metric	Consolidated Value
Avg Fidelity	89.5%
Pipeline Status	Advanced Test

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Dataset Audit: demo_data.csv

1. Dataset Portrait

Attribute	Value
Rows	100
Cols	2
Missing	0
Memory	12 KB
Target Variables	A

2. Generation Strategy & LLM Config

Parameter	Configuration
Algorithm	gemma-3-4b
Style	Formal
Consistency	Strict
Temperature	0.7
Chunk Size	100
Max Context	4096
Strategy	Diverse

3. Statistical Significance & Data Drift

Aggregated across all k-folds. Drift defined as KS-Test p-value < 0.05.

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
A	0.01	0.02	+1.2%	0.8421	Pass
B	-0.12	-0.10	-1.5%	0.9122	Pass

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Score (aggregated)
Fidelity Score	91.2 ± 2.1
Privacy Risk	5.4 ± 0.3

5. Downstream ML Utility Results

Model / Metric	Aggregated Performance
F1 Score	0.8842
Accuracy	0.8911

Appendix: Multi-Fold Data Trace

Sample rows from original and each generated fold for audit trailing.

Baseline: Original Data Sample

A	B
-0.347995815745262	1.1047519048816463
1.1183595706661367	1.2344131817666482
-0.5795694361839646	0.9687695036433079
-0.4897251593203574	-0.13722330788252035
-0.23807848631489717	-0.7570135098809528

Audit Trace: Fold 1

A	B
-0.347995815745262	1.1047519048816463
1.1183595706661367	1.2344131817666482
-0.5795694361839646	0.9687695036433079
-0.4897251593203574	-0.13722330788252035
-0.23807848631489717	-0.7570135098809528

Audit Trace: Fold 2

A	B
-0.347995815745262	1.1047519048816463
1.1183595706661367	1.2344131817666482
-0.5795694361839646	0.9687695036433079
-0.4897251593203574	-0.13722330788252035
-0.23807848631489717	-0.7570135098809528